

Calculation of Lift and Drag from Surface Pressure Measurements

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The purpose of this experiment was to find the lift and drag acting on two airfoils, NACA 0012 and NACA 4412, by measuring the pressure distributions around both airfoils at multiple angles of attack. The pressure coefficients C_p were found using a scanivalve connected to multiple pressure taps along the upper and lower surfaces of both NACA airfoils then used the dynamic and static pressure of the test section measured with a pitot tube system. Using this data the lift and drag coefficients C_L and C_D were computed in MATLAB using integration of pressure distributions. All wind tunnels measurements are conducted at a constant fan frequency of 40 Hz while the airfoils were gradually rotating between each test to capture the aerodynamic values. Doing so resulted in the symmetric NACA 0012 producing zero lift when at an angle of attack of 0 degrees while the cambered NACA 4412 gave off a positive lift at 0 degrees. These results prove the influence of camber and angle of attack on airfoils

I. Nomenclature

C_p	=	pressure coefficient
p	=	static pressure (Pa)
p_∞	=	freestream static pressure (Pa)
q_∞	=	freestream dynamic pressure (Pa)
C_z	=	normal-force coefficient
C_x	=	axial-force coefficient
C_L	=	lift coefficient
C_D	=	drag coefficient
L	=	lift force (N)
D	=	drag force (N)
c	=	airfoil chord length (m)
q	=	dynamic pressure (Pa)
P_{sca}	=	pressure measured by scanivalve (Pa)
δP_{sca}	=	uncertainty in scanivalve pressure (Pa)
δq_∞	=	uncertainty in dynamic pressure (Pa)
δC_p	=	uncertainty in pressure coefficient
$C_{L,max,C}$	=	maximum and minimum lift coefficient
$C_{D,max,C}$	=	maximum and minimum drag coefficient
δC_L	=	uncertainty in lift coefficient
δC_D	=	uncertainty in drag coefficient
α	=	angle of attack
ρ	=	air density

II. Introduction

Wind tunnels are aerodynamic devices that use high-speed fans to move air through a test section to study the effects of airflow under specific conditions and around different-shaped bodies such as the NACA airfoils. In this experiment the wind tunnels help create controlled flow conditions and allow for the measurement of pressure distributions to help determine lift, drag, and pressure coefficients.

The objective of this experiment was to determine the lift and drag acting on the two airfoils, NACA 0012 and NACA 4412, by measuring the pressure distributions along the surface at different angles of attack. In the experiment, the measured pressures were used to find the pressure coefficient C_p , and the values were then integrated to obtain the lift and drag coefficients C_l and C_d . Next, the results were used to find the difference between symmetric and cambered airfoils at different angles of attack.

The main devices used in this lab experiment include the scannivalve, pitot tube system, wind tunnel, and the airfoil that intakes the gauge pressure from the pressure ports connected to the scannivalve. The pitot tube provided dynamic and static pressures used to find the pressure coefficient at all airfoil ports. Using all the tools allowed for the data required to find the pressure coefficient at all airfoil ports and angles of attack.

The analysis of the lab was based on Bernoulli's equation with the assumptions of incompressible, inviscid, steady-state, and uniform flow. This relates the pressure and velocity on a streamline and allows the pressure difference over the airfoil surface to be found. Using the integration method in a clockwise direction around the airfoil geometry, the pressure data was used to find the lift and drag coefficients. With these important values found, you can determine important insights based on the angle of attack of both NACA airfoils.

III. Procedure

Start experiment by recording the ambient temperature and pressure of the laboratory using a calibrated thermometer and barometer. Next attach the NACA 0012 airfoil to the wind tunnel from wall to wall to replicate a two-dimensional flow field. Connect the nine airfoil ports with port one being the leading edge and increasing by one until the ninth port at the trailing edge. attach to the scanivalve using pressure tubing with port 1 corresponding to the leading edge and 9 to the trailing edge for the NACA 0012 airfoil. Attach one pressure tube to the front of the test section floor so the scannivalve will measure the gauge pressure around the airfoil. Finally ensure the pressure transducer is connected across the pitot static tube to record the test sections dynamic pressure. The experiment setup can be seen below.

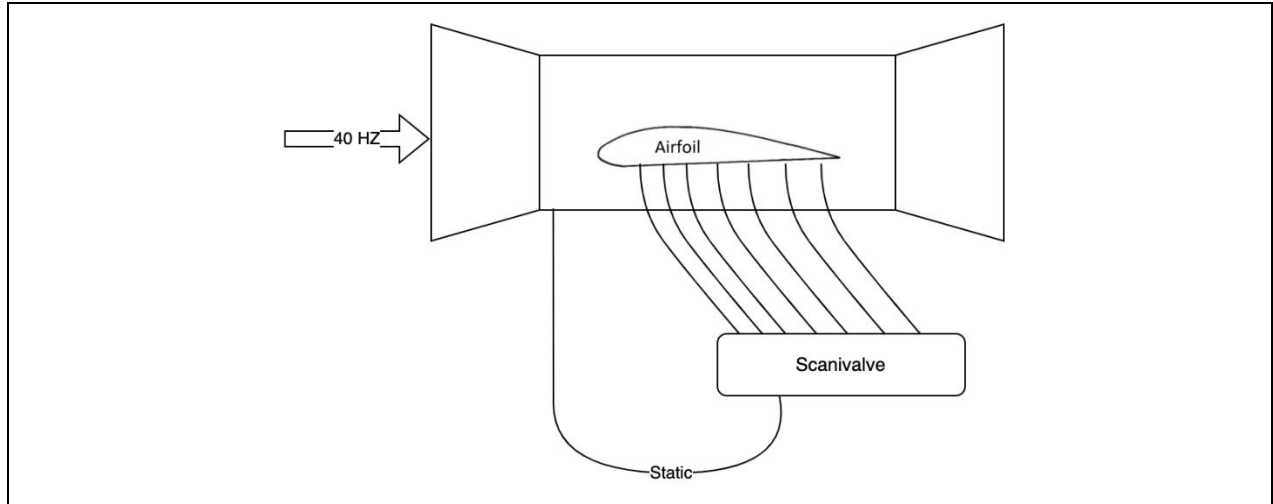


Fig. 1 Diagram of the wind tunnel setup

Begin testing with the NACA 0012 airfoil at an angle of attack of 0° and turn on the wind tunnel to a constant speed of 40 Hz. Measure the static pressure at each port by twisting the knob on the scanivalve from port 1 to 9, recording the data at each point. After all ports are recorded, adjust the angle of attack (0° , 4° , 8° , 10° , 12°) until all data is recorded for the top surface. Because NACA 0012 is symmetrical, repeat the measurements for the bottom surface by using negative angles of attack (0° , -4° , -8° , -10° , -12°). Now swap out the NACA 0012 with the cambered NACA 4412 airfoil and begin by attaching the pressure tubing to the top ports 1-8 to record the top surface. Again, record data at angles of attack (0° , 4° , 8° , 10° , and 14°) for each scanivalve port, then change the pressure tubing to the lower surface ports 1-7 and record at the same angles of attack.

After all pressure data are obtained, compute the surface pressure coefficients using (Eq. (1)). The pressure coefficients should be found at every pressure tap, upper and lower, and plotted vs. the chordwise for all angles of attack, resulting in ten graphs.

$$C_p = \frac{p - p_\infty}{q_\infty} \quad (1)$$

The components of pressure force coefficients were then computed through trapezoidal integration of the calculation of the pressure coefficient distribution around the airfoil's surface given in the equations (Eq. (2)-(3)).

$$C_z = - \int C_p dx \quad (2)$$

$$C_x = \int C_p dz \quad (3)$$

Next, the components can be changed into lift and drag directions based on the angle of attack (Eq. (4)-(5)).

$$C_L = C_z \cos \alpha - C_x \sin \alpha \quad (4)$$

$$C_D = C_x \cos \alpha + C_z \sin \alpha \quad (5)$$

The lift and drag coefficients C_L and C_D dimensionless values that represent the amount of lift and drag an airfoil may generate and are used to determine lift and drag (Eq. (6)-(7)).

$$L = C_L q_\infty c \quad (6)$$

$$D = C_D q_\infty c \quad (7)$$

To determine the measurement precision, use the known uncertainty from the laboratory instrumentation, the transducer, and the scanivalve, which can be used to calculate the pressure coefficient. Convert the voltage uncertainties into pressure uncertainties by applying the calibration slopes derived from the manometer-to-voltage fits. Then compute the uncertainty in the pressure coefficient using (Eq. (8)).

$$\delta C_p = \sqrt{\left(\frac{\partial C_p}{\partial P_{sca}} \delta P_{sca}\right)^2 + \left(\frac{\partial C_p}{\partial q_\infty} \delta q_\infty\right)^2} \quad (8)$$

Then represent each found point of data as (Eq. (13))

$$C_p \pm \delta C_p \quad (9)$$

Finally, to determine the uncertainties in the integrated lift and drag coefficients, repeat the integration using the upper and lower bound C_p distributions. Every angle of attack calculates C_L and C_D using the nominal, maximum, and minimum pressure coefficients. Then evaluate the spread between the maximum and minimum integrated values to calculate the uncertainties (Eq. (10)-(11)).

$$\delta C_L = \frac{C_{L,max} - C_{L,min}}{2} \quad (10)$$

$$\delta C_D = \frac{C_{D,max} - C_{D,min}}{2} \quad (11)$$

This provides the nominal values and the propagated measurement uncertainties of lift and drag coefficients, which provide accurate precision.

IV. Results

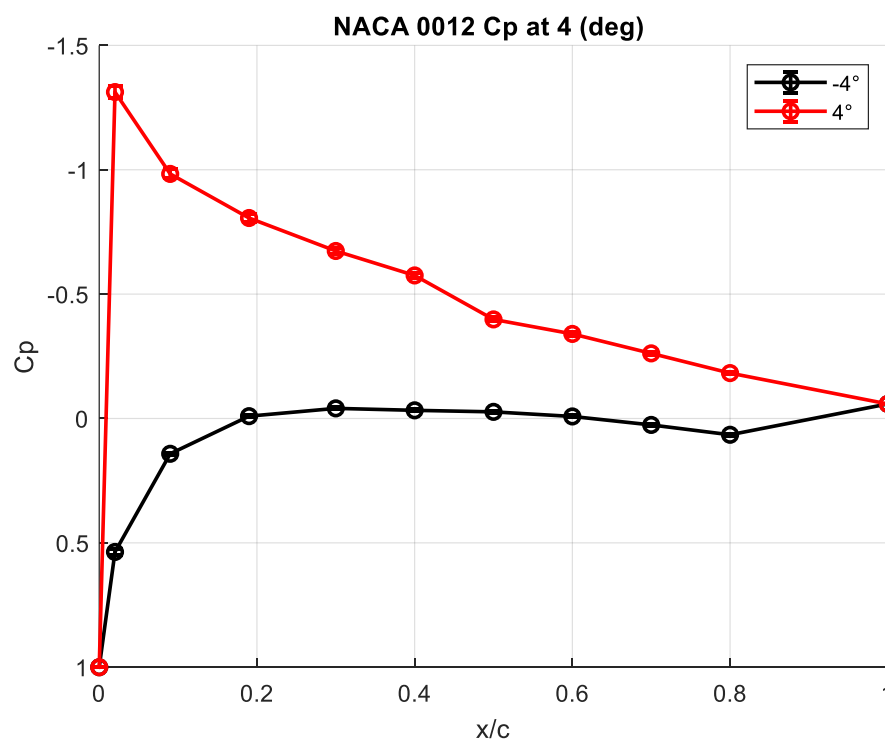
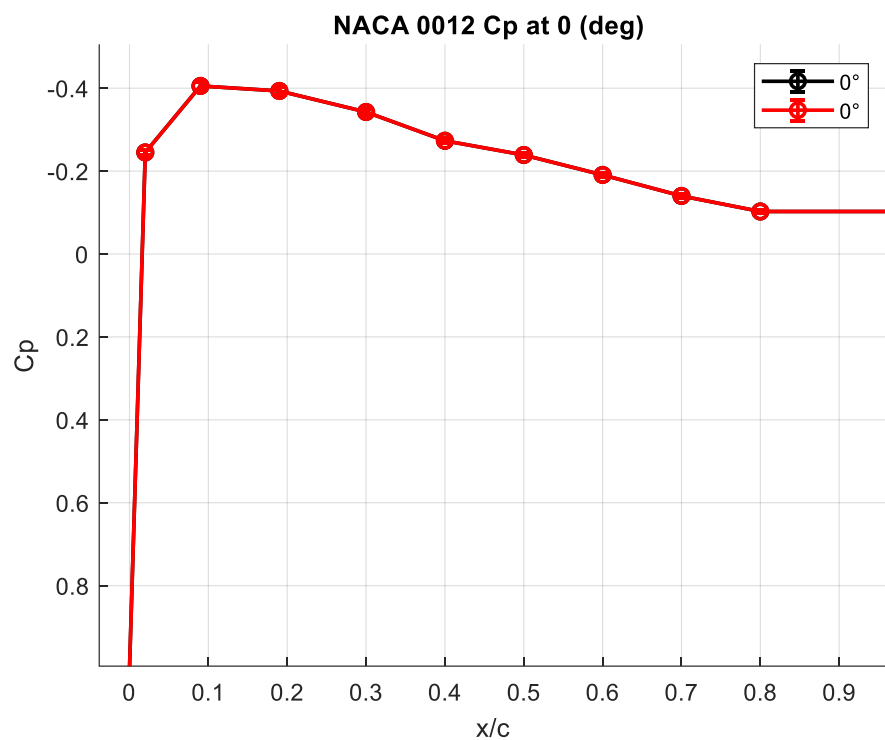
Fig. 1-10 C_p vs x/c displays the pressure coefficients C_p distributions for each angle of attack. NACA 0012 upper and lower surface pressure distributions were identical, proving that a symmetric airfoil produces no lift when directed into the flow. As the angle of attack increased, the upper surface had lower pressure, while the pressure on the lower surface slightly increased. This creates a greater pressure difference, resulting in high lift until at the point where the pressure difference decreases again due to stalling. For the NACA 4412 airfoil, the camber causes the upper and lower surfaces to separate, which generates lift. As the angle increases, the top surface decreases in pressure and even extends throughout the chord length. Looking at the lower surface, there is a very small change; this shows that the lift generated from the cambered airfoil comes from low pressure due to the top surface.

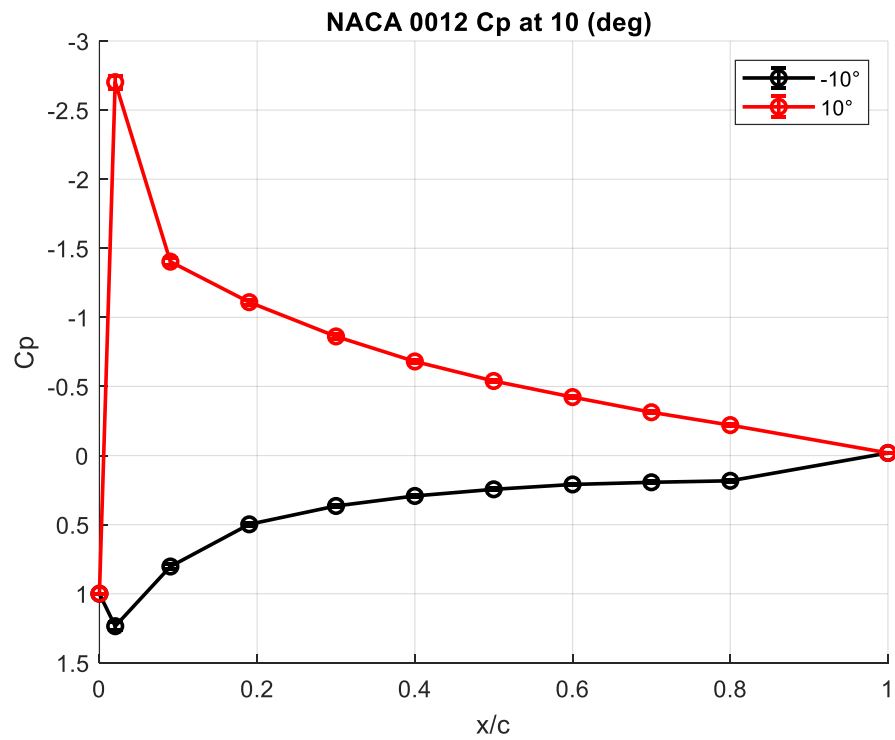
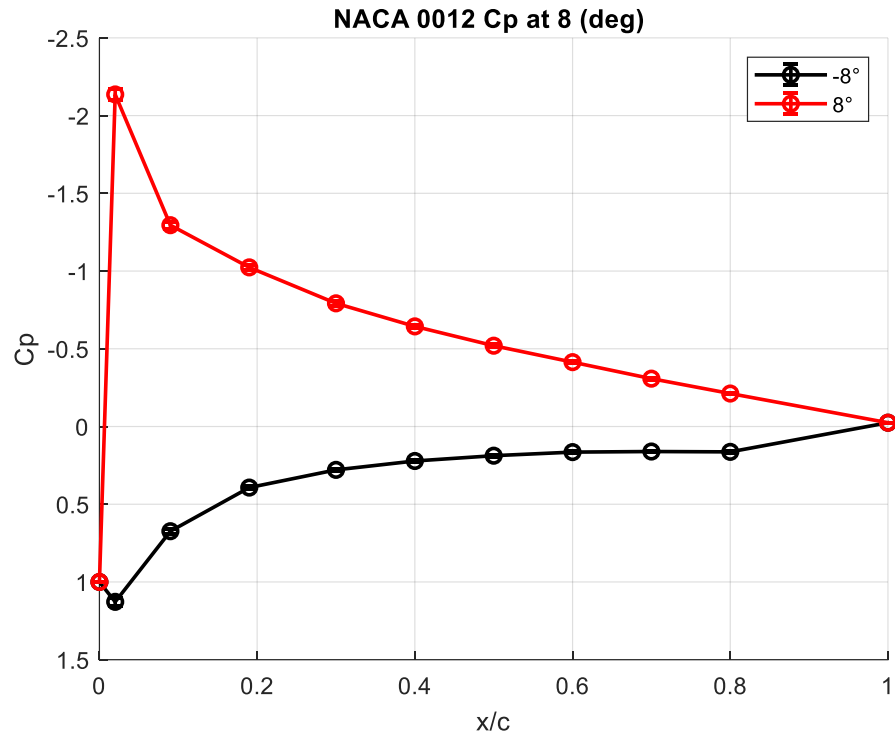
For NACA 0012 (Fig. 11), the lift coefficient increased constantly until an attack of 10° matching with thin air foil theory before decreasing due to stall. The drag coefficient remained small until passing the stall point due to the separation of flows at higher angles of attack. The L/D ratio is low at 0° , showing the symmetric shape generates no lift without an angle of attack since there isn't a big pressure difference. NACA 4412 (Fig. 12), lift is instantly positive at 0° because of the cambered shape, which also maintains high lift after the stall angle. The drag coefficient increased slowly with the angle of attack but remains smaller than the symmetric airfoil. This shows the importance of camber for a successful airfoil.

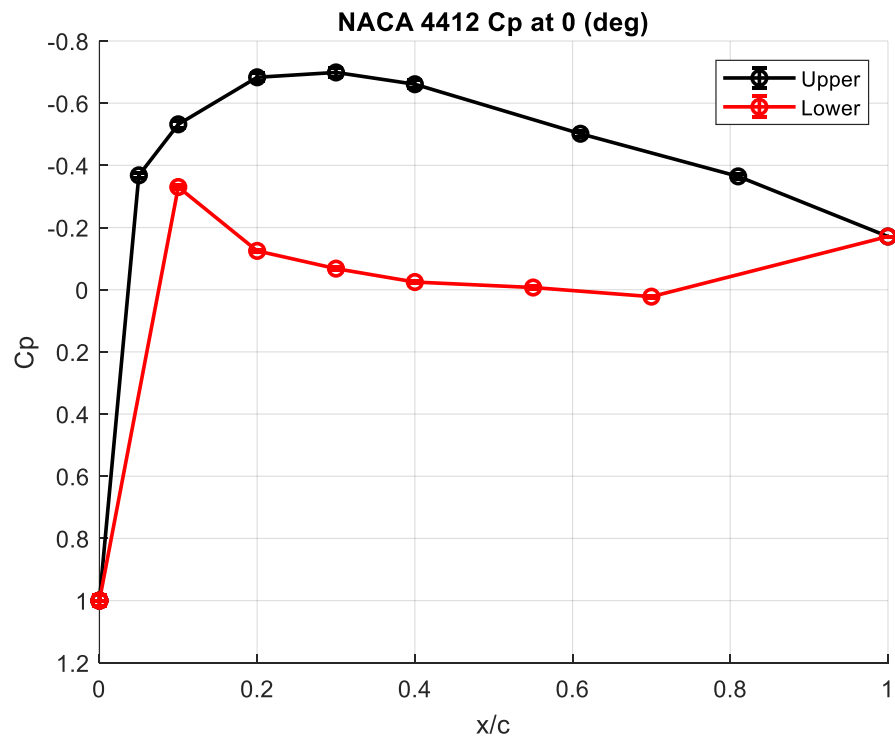
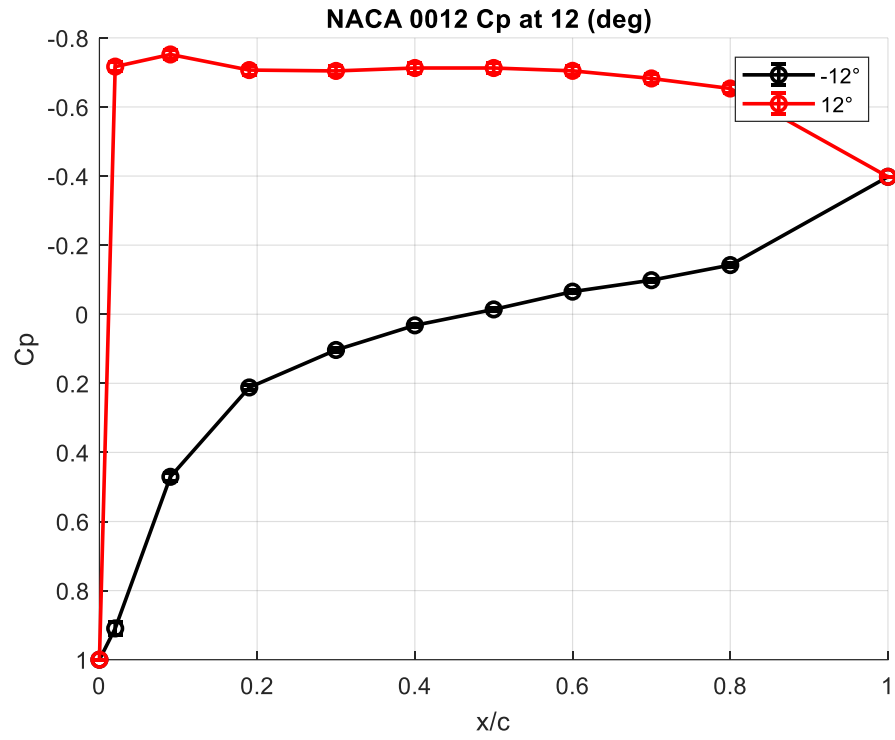
Figures 13 and 14 present the uncertainty bounds for both NACA airfoils. For the NACA 0012, lift and drag values are closely bounded at all angles with a slight spread near the stall angle of 10° . For the NACA 4412, uncertainty bounds are similar, with both airfoils having a tight C_D bound.

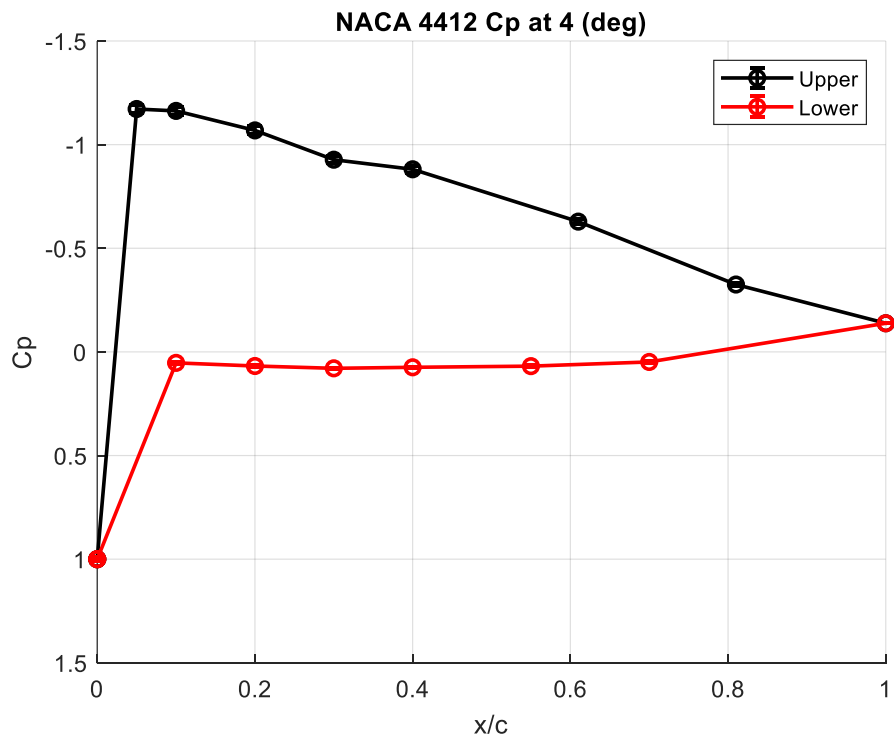
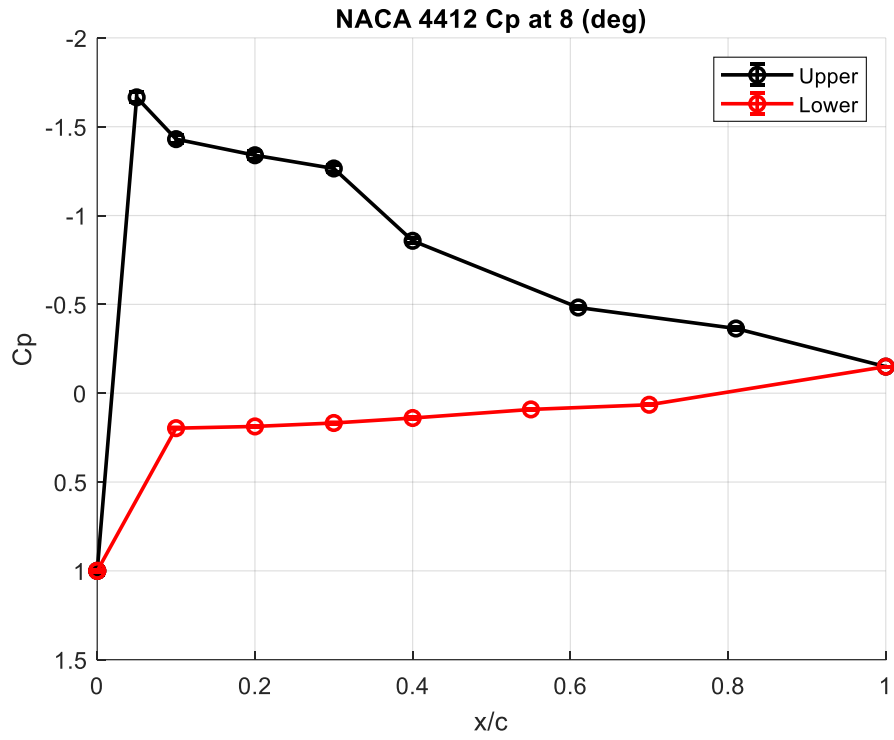
The estimated instrument uncertainties, ($\delta V_{transducer} = 0.01$ V, $\delta V_{scanivalve} = 0.005$ V) from machine errors only gave off small pressure errors δC_p less than ± 0.02 . The lift and drag coefficient uncertainties were determined from the upper and lower bound C_p distributions using (Eq. (10)-(11)), giving $\delta C_L \approx \pm 0.03$ and giving $\delta C_D \approx \pm 0.01$. These results match (Fig. 11 and 12), as the lower surface is tighter to C_D and slightly looser for C_L .

Fig. 1-10 Pressure coefficient ($-C_p$) vs angle of attack









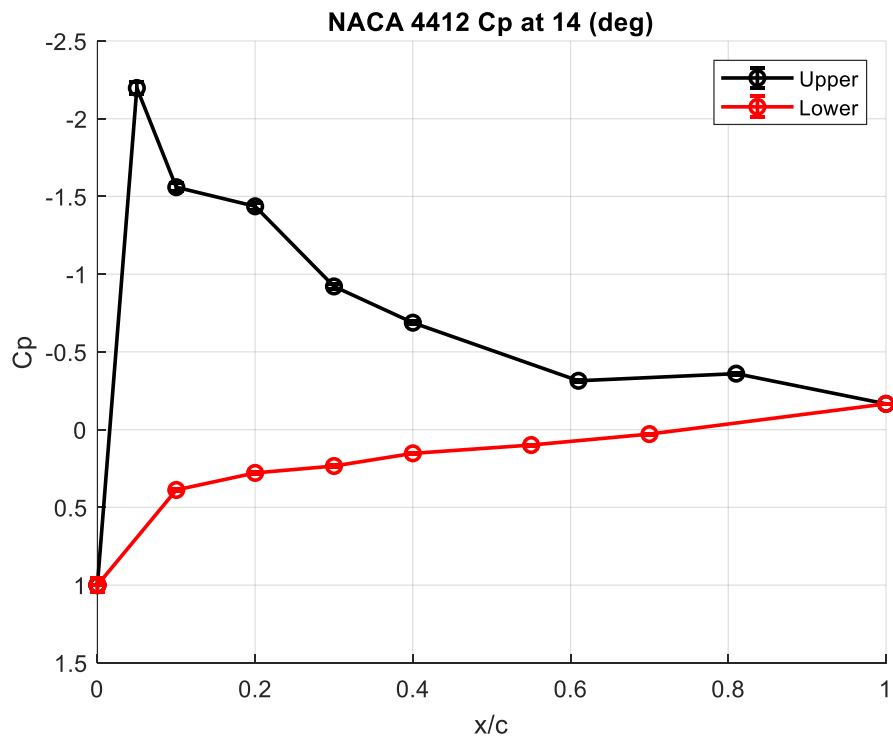
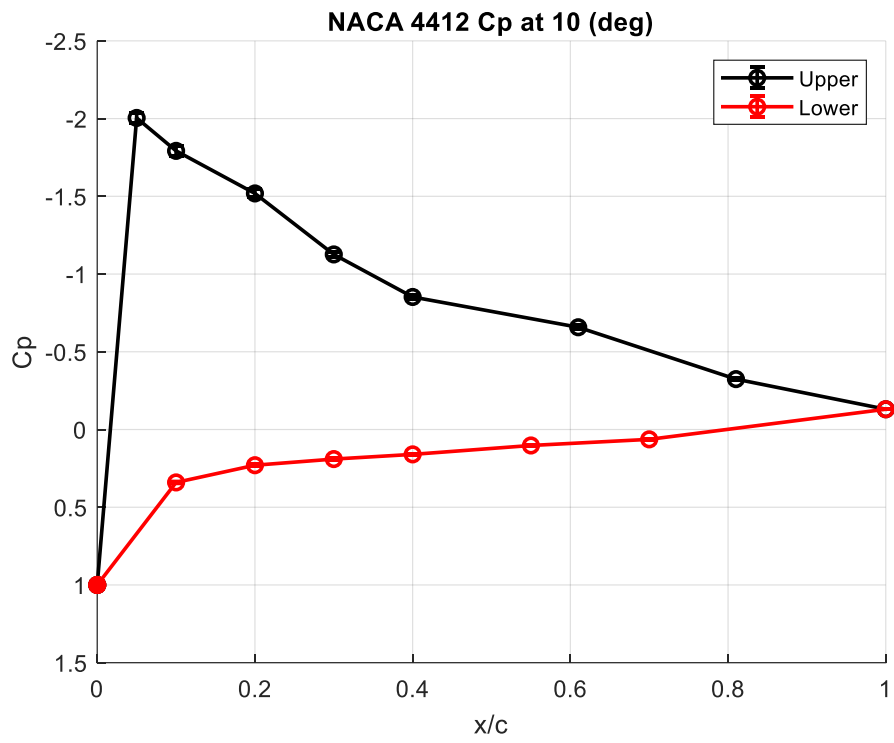


Fig. 11-12: C_D , and L/D vs angle of attack

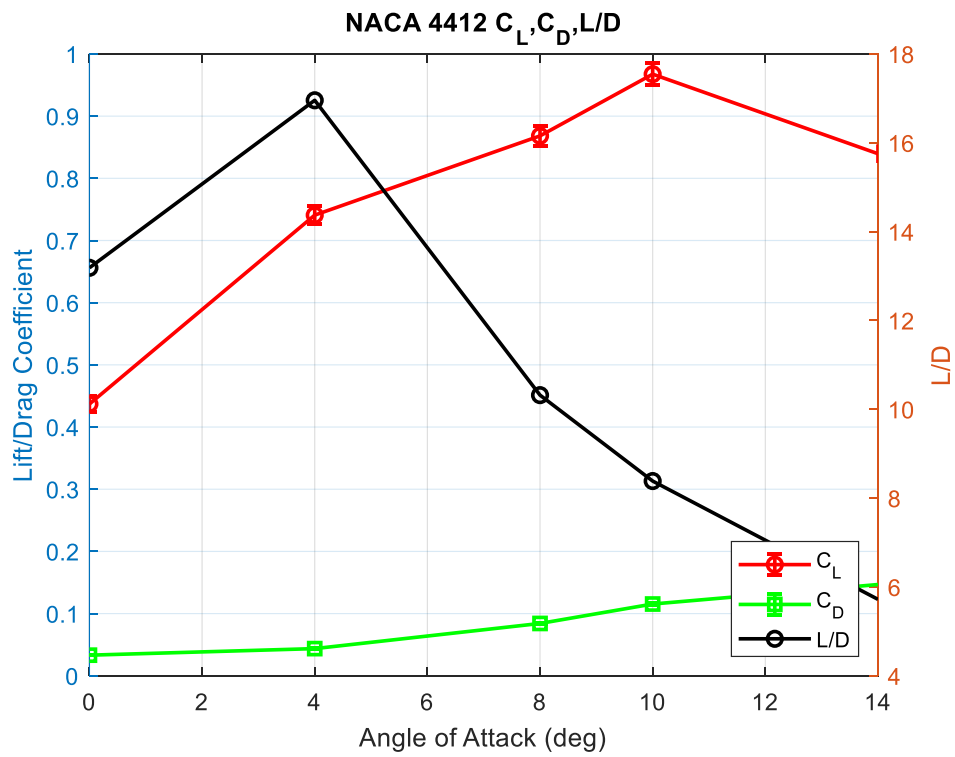
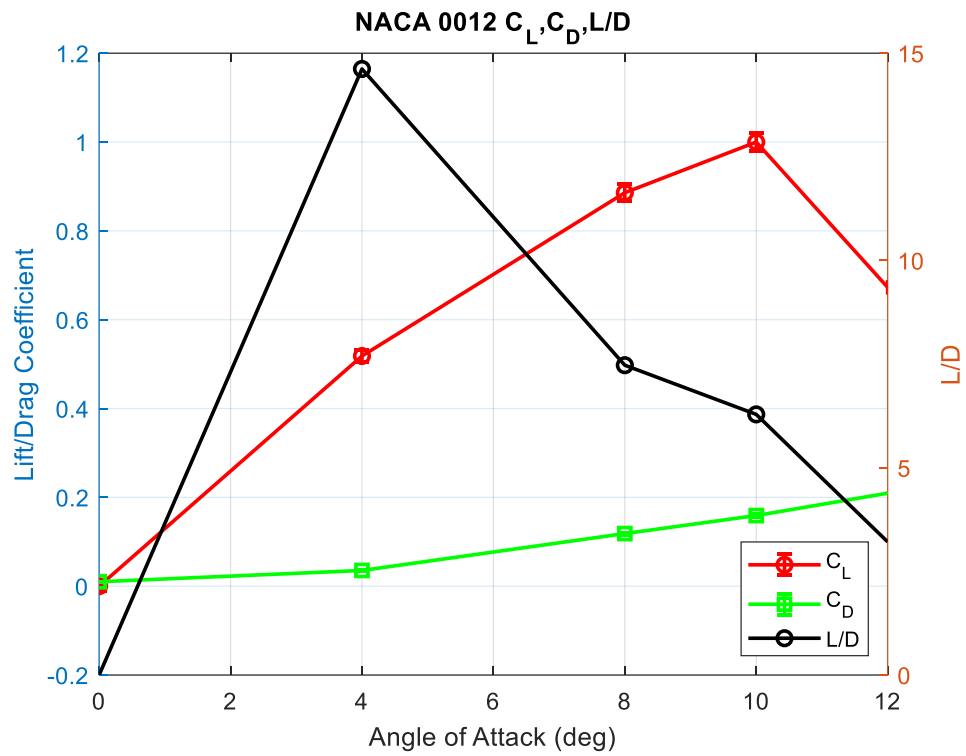
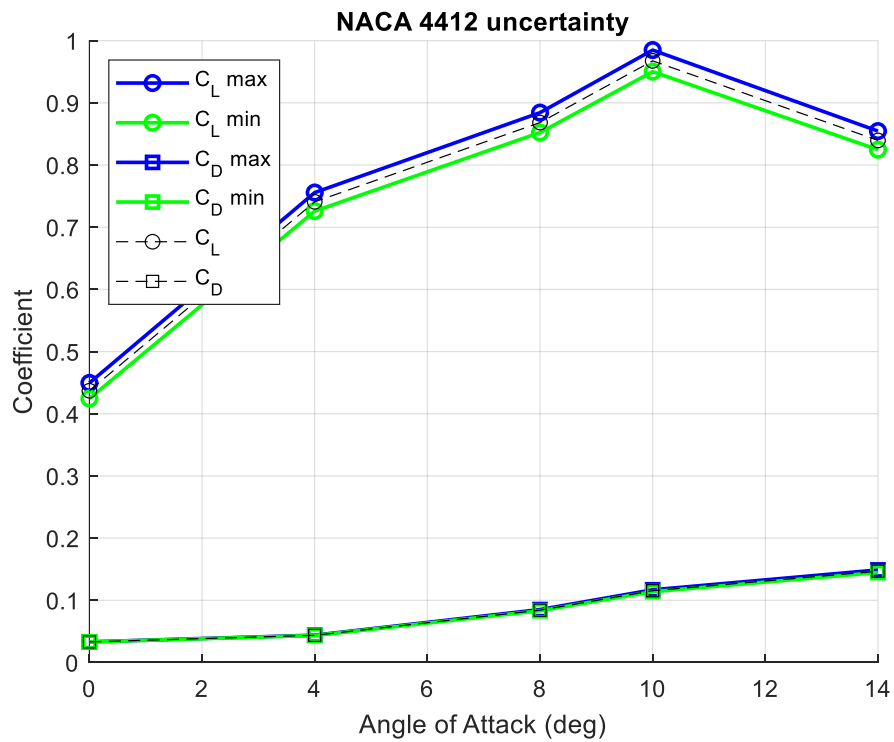
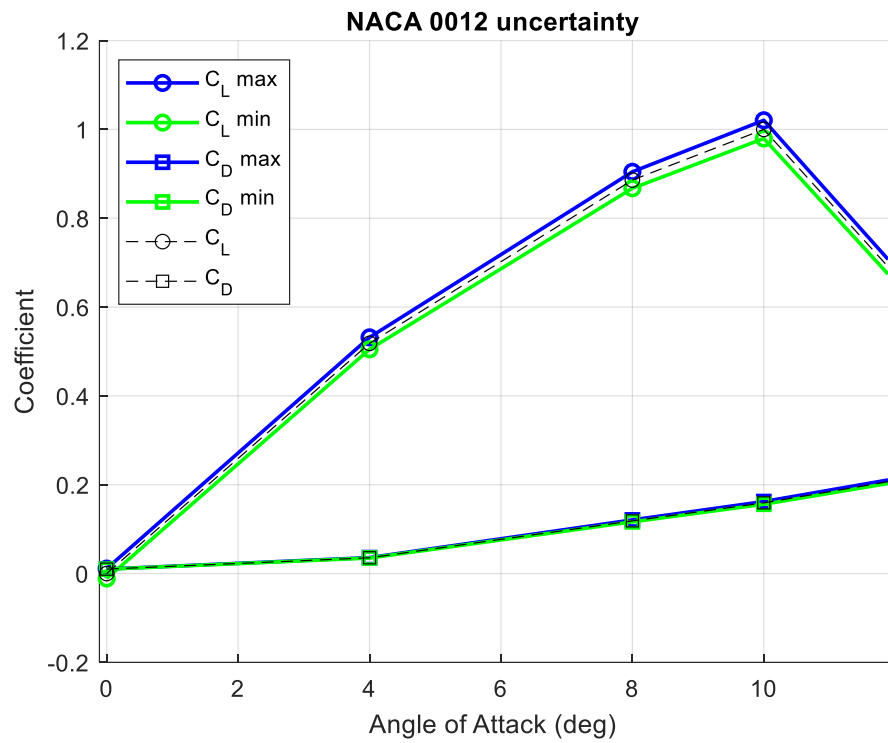


Fig. 13-14: uncertainty bounds $C_{L,max}$, $C_{L,min}$, $C_{D,max}$, and $C_{D,min}$ vs angle of attack.



V. Conclusion

The collected results of this aerodynamic experiment aligned closely with expectations. NACA 0012, the symmetric airfoil, produced zero lift at zero angle of attack and followed a linear slope to 10° matching thin airfoil theory as predicted. The NACA 4412 produces lift at an angle of attack of 0° and has greater lift-to-drag ratios than the symmetric airfoil. Both airfoils displayed a stall region when the angle of attack was too steep and the flow separated.

The experiment's uncertainty came out tiny, meaning that the results were accurate and able to be repeated for future tests. Things that could contribute to having small discrepancies include possible air leaks on connection ports, no pressure port on the trailing edge, and machine error. Possible solutions to make the experiment results more accurate include using more pressure ports to receive data across the whole airfoil since our tests are lacking a trailing edge port and having more accurate instruments to gather more precise data.

Overall, the experiment demonstrates the connection between pressure distributions and their aerodynamic effects, such as lift and drag. When comparing the symmetric and cambered airfoils, they showed how airfoil geometry affects lift and drag. Therefore, this experiment proved important concepts of aerodynamics like the lift-curve slope, airfoil theory, stall behavior, and the importance of airfoil shape. In summary, this experiment was very important because it applied useful aerodynamic properties and gave a connection between theory and how real aircraft wings are designed for high lift and low drag for max efficiency.

VI. Appendix

Appendix A: Tabulated Data

Table 1: NACA 0012 upper surface C_p with uncertainty

Angle (°)	LE	0.02	0.09	0.19	0.3	0.4	0.5	0.6	0.7	0.8	TE
0°	1.000 ±0.000	-0.245 ±0.006	-0.405 ±0.009	-0.393 ±0.009	-0.343 ±0.008	-0.273 ±0.006	-0.239 ±0.006	-0.191 ±0.005	-0.140 ±0.005	-0.103 ±0.004	-0.103 ±0.003
4°	1.000 ±0.000	-1.312 ±0.024	-0.983 ±0.018	-0.806 ±0.015	-0.673 ±0.013	-0.575 ±0.011	-0.398 ±0.008	-0.340 ±0.007	-0.262 ±0.006	-0.183 ±0.005	-0.059 ±0.003
8°	1.000 ±0.000	-2.136 ±0.037	-1.295 ±0.023	-1.024 ±0.018	-0.792 ±0.014	-0.644 ±0.012	-0.520 ±0.010	-0.414 ±0.008	-0.308 ±0.006	-0.212 ±0.005	-0.024 ±0.004
10°	1.000 ±0.000	-2.702 ±0.046	-1.403 ±0.024	-1.110 ±0.019	-0.862 ±0.015	-0.681 ±0.012	-0.539 ±0.010	-0.423 ±0.008	-0.313 ±0.006	-0.220 ±0.005	-0.019 ±0.004
12°	1.000 ±0.000	-0.717 ±0.013	-0.752 ±0.014	-0.707 ±0.013	-0.704 ±0.013	-0.713 ±0.013	-0.713 ±0.013	-0.705 ±0.013	-0.682 ±0.013	-0.653 ±0.012	-0.398 ±0.007

Table 2: NACA 0012 lower surface C_p with uncertainty

Angle (°)	LE	0.02	0.09	0.19	0.3	0.4	0.5	0.6	0.7	0.8	TE
0°	1.000 ±0.000	-0.245 ±0.006	-0.405 ±0.009	-0.393 ±0.009	-0.343 ±0.008	-0.273 ±0.006	-0.239 ±0.006	-0.191 ±0.005	-0.140 ±0.005	-0.103 ±0.004	-0.103 ±0.003
-4°	1.000 ±0.000	0.537 ±0.012	0.142 ±0.005	-0.009 ±0.004	-0.041 ±0.004	-0.033 ±0.004	-0.027 ±0.004	-0.008 ±0.004	0.026 ±0.004	0.065 ±0.004	-0.059 ±0.003
-8°	1.000 ±0.000	1.128 ±0.026	0.674 ±0.016	0.393 ±0.010	0.279 ±0.007	0.222 ±0.006	0.188 ±0.006	0.165 ±0.006	0.161 ±0.005	0.163 ±0.006	-0.024 ±0.004
-10°	1.000 ±0.000	1.235 ±0.028	0.803 ±0.019	0.498 ±0.012	0.365 ±0.009	0.292 ±0.008	0.244 ±0.007	0.209 ±0.006	0.194 ±0.006	0.183 ±0.006	-0.019 ±0.004
-12°	1.000 ±0.000	0.910 ±0.020	0.471 ±0.011	0.212 ±0.006	0.103 ±0.005	0.032 ±0.004	-0.014 ±0.004	-0.065 ±0.004	-0.098 ±0.004	-0.142 ±0.005	-0.398 ±0.007

Table 3: NACA 4412 upper surface C_p with uncertainty

Angle (°)	LE	0.05	0.1	0.2	0.3	0.4	0.61	0.81	TE
0°	1.000 ±0.000	1.000 ±0.017	-0.368 ±0.008	-0.532 ±0.011	-0.683 ±0.013	-0.699 ±0.014	-0.661 ±0.013	-0.502 ±0.010	-0.365 ±0.008
4°	1.000 ±0.000	1.000 ±0.005	-1.172 ±0.021	-1.163 ±0.021	-1.069 ±0.019	-0.927 ±0.017	-0.881 ±0.016	-0.628 ±0.011	-0.325 ±0.007
8°	1.000 ±0.000	1.000 ±0.024	-1.665 ±0.028	-1.430 ±0.024	-1.339 ±0.023	-1.265 ±0.021	-0.858 ±0.015	-0.482 ±0.009	-0.364 ±0.007
10°	1.000 ±0.000	1.000 ±0.028	-2.004 ±0.033	-1.792 ±0.030	-1.519 ±0.025	-1.126 ±0.019	-0.853 ±0.014	-0.658 ±0.011	-0.324 ±0.006
14°	1.000 ±0.000	1.000 ±0.046	-2.198 ±0.035	-1.559 ±0.025	-1.437 ±0.023	-0.921 ±0.015	-0.688 ±0.011	-0.314 ±0.006	-0.360 ±0.007

Table 4: NACA 0012 lower surface Cp with uncertainty

Angle (°)	LE	0.1	0.2	0.3	0.4	0.55	0.7	TE	TE
0°	1.000 ±0.000	1.000 ±0.018	-0.330 ±0.007	-0.125 ±0.004	-0.068 ±0.004	-0.025 ±0.004	-0.007 ±0.004	0.022 ±0.004	-0.171 ±0.004
4°	1.000 ±0.000	1.000 ±0.009	0.053 ±0.003	0.068 ±0.003	0.079 ±0.004	0.074 ±0.004	0.069 ±0.003	0.049 ±0.003	0.049 ±0.003
8°	1.000 ±0.000	1.000 ±0.006	0.197 ±0.005	0.187 ±0.004	0.168 ±0.004	0.140 ±0.004	0.091 ±0.003	0.065 ±0.003	0.065 ±0.003
10°	1.000 ±0.000	1.000 ±0.027	0.340 ±0.006	0.230 ±0.005	0.190 ±0.004	0.160 ±0.004	0.103 ±0.003	0.063 ±0.003	0.063 ±0.003
14°	1.000 ±0.000	1.000 ±0.045	0.388 ±0.007	0.278 ±0.005	0.234 ±0.005	0.153 ±0.004	0.099 ±0.003	0.029 ±0.003	0.029 ±0.003

Table 5: NACA 0012 CL, CD, L/D with uncertainty

Angle (°)	C _L	C _D ±	L/D
0°	0.0000 ±0.0111	0.0100 ±0.0001	0
4°	0.5183 ±0.0134	0.0354 ±0.0005	14.62
8°	0.8859 ±0.0186	0.1185 ±0.0022	7.47
10°	1.0000 ±0.0205	0.1590 ±0.0031	6.29
12°	0.6729 ±0.0163	0.2095 ±0.0040	3.21

Table 6: NACA 0012 CL, CD, L/D with uncertainty

Angle (°)	C _L	C _D	L/D
0°	0.4370 ±0.0126	0.0331 ±0.0001	13.19
4°	0.7408 ±0.0149	0.0437 ±0.0002	16.96
8°	0.8683 ±0.0160	0.0841 ±0.0010	10.32
10°	0.9675 ±0.0173	0.1154 ±0.0015	8.38
14°	0.8396 ±0.0150	0.1468 ±0.0021	5.72

Appendix B: MATLAB Code

CODE:

```
clc; clear; close all;

%% AEE 360 Lab 2

% Files
file0012 = 'Fri_B_730_0012.txt';
file4412_U = 'Fri_B_730_4412u.txt';
file4412_L = 'Fri_B_730_4412l.txt';

% Constants & uncertainty from Lab 1
INH2O_TO_PA = 248.84; % in of H2O to Pa
Pa_transducer = 1336.5; % Pa/V
Pa_scanivalve = 493.5; % Pa/V
dV_transducer = 0.01; % V
dV_scanivalve = 0.005; % V
dq_inf = dV_transducer * Pa_transducer;
dp_scan = dV_scanivalve * Pa_scanivalve;

% Geometry
xC_0012_ports = [2 9 19 30 40 50 60 70 80]/100;
xC_4412_U = [0 0.05 0.10 0.20 0.30 0.40 0.61 0.81];
xC_4412_L = [0 0.10 0.20 0.30 0.40 0.55 0.70];

zC_4412_U = [0 0.0449 0.0643 0.0874 0.0975 0.0980 0.0797 0.0458];
zC_4412_L = [0 -0.0293 -0.0274 -0.0225 -0.0180 -0.0119 -0.0063];

x0012 = xC_0012_ports;
t0012 = (0.12/0.2).*( 0.2969*sqrt(x0012) - 0.1260*x0012 ...
    - 0.3516*x0012.^2 + 0.2843*x0012.^3 - 0.1015*x0012.^4 );
zU_0012 = t0012;
zL_0012 = -t0012;

%% Load data
raw0012 = importdata(file0012);
raw4412_U = importdata(file4412_U);
raw4412_L = importdata(file4412_L);

% data to rows
q0012 = raw0012.data(:,3);
aoa0012 = raw0012.data(:,4);
inH2O0012 = raw0012.data(:,5);
port0012 = raw0012.data(:,6);
q4412_U = raw4412_U.data(:,3);
aoa4412_U = raw4412_U.data(:,4);
inH2O4412_U = raw4412_U.data(:,5);
```

```

port4412_U = raw4412_U.data(:,6)';
q4412_L = raw4412_L.data(:,3)';
aoa4412_L = raw4412_L.data(:,4)';
inH2O4412_L = raw4412_L.data(:,5)';
port4412_L = raw4412_L.data(:,6)';

% NACA 0012 Cp and uncert Cp
angles12 = unique(aoa0012);
nPorts12 = numel(xC_0012_ports);
Cp12 = zeros(numel(angles12), nPorts12);
dCp12 = zeros(numel(angles12), nPorts12);

for i = 1:numel(angles12)
    a = angles12(i);
    for p = 1:nPorts12
        hit = (aoa0012==a) & (port0012==p);
        if any(hit)
            p_port = inH2O0012(hit) * INH2O_TO_PA;
            qinf = q0012(hit);
            Cp12(i,p) = p_port ./ qinf;
            dCp12(i,p) = sqrt( (dp_scan./qinf).^2 + ((-p_port./(qinf.^2)).*dq_inf).^2 );
        end
    end
end

% NACA 0012 Cp plots
pairs12 = {[0,0],[-4,4],[-8,8],[-10,10],[-12,12]};
for k = 1:numel(pairs12)
    aN = pairs12{k}(1);
    aP = pairs12{k}(2);
    cpN = Cp12(angles12==aN,:);
    dN = dCp12(angles12==aN,:);
    cpP = Cp12(angles12==aP,:);
    dP = dCp12(angles12==aP,:);
    if isempty(cpN) || isempty(cpP), continue;
    end
    cpTE = 0.5*(cpP(end)+cpN(end));
    xle = [0 xC_0012_ports 1];
    figure; hold on;
    errorbar(xle, [1 cpN cpTE], [0 dN 0], 'ko-', 'LineWidth', 1.5, 'DisplayName', sprintf('%d°', aN));
    errorbar(xle, [1 cpP cpTE], [0 dP 0], 'ro-', 'LineWidth', 1.5, 'DisplayName', sprintf('%d°', aP));
    xlabel('x/c');
    ylabel('Cp');
    title(sprintf('NACA 0012 Cp at %d (deg) ', aP));
    legend;
    grid on;
    axis ij;
end

```



```

% NACA 0012 Integrate CL and CD
x_Loop12 = [0, xC_0012_ports, 1, xC_0012_ports(end:-1:1), 0];
z_Loop12 = [0, zU_0012, 0, zL_0012(end:-1:1), 0];

CL12 = zeros(numel(angles12),1);
CD12 = zeros(numel(angles12),1);
CL12_min = CL12;
CL12_max = CL12;
CD12_min = CD12;
CD12_max = CD12;

for i = 1:numel(angles12)
    a = angles12(i);

    CpU_true = Cp12(angles12==a,:);
    CpL_true = Cp12(angles12==a,:);
    if isempty(CpL_true), CpL_true = CpU_true;
    end

    % four combinations for bounds
    Umin = CpU_true - dCp12(angles12==a,:);
    Umax = CpU_true + dCp12(angles12==a,:);
    if any(angles12==a)
        Lmin = CpL_true - dCp12(angles12==a,:);
        Lmax = CpL_true + dCp12(angles12==a,:);
    else
        Lmin = CpL_true - dCp12(angles12==a,:);
        Lmax = CpL_true + dCp12(angles12==a,:);
    end
    combos = { {Umin,Lmin}, {Umin,Lmax}, {Umax,Lmin}, {Umax,Lmax} };

    CLs = zeros(1,4);
    CDs = zeros(1,4);
    for c = 1:4
        Uc = combos{c}{1}; Lc = combos{c}{2};
        cpTE = 0.5*(Uc(end)+Lc(end));
        cpLoop = [1, Uc, cpTE, Lc(end:-1:1), 1];
        Fx = trapz(z_Loop12, cpLoop);
        Fz = trapz(x_Loop12, -cpLoop);
        CLs(c) = Fz*cosd(a) - Fx*sind(a);
        CDs(c) = Fz*sind(a) + Fx*cosd(a);
    end

    cp_TEt = 0.5*(CpU_true(end)+CpL_true(end));
    cp_Loop_T = [1, CpU_true, cp_TEt, CpL_true(end:-1:1), 1];
    Fx_T = trapz(z_Loop12, cp_Loop_T);
    Fz_T = trapz(x_Loop12, -cp_Loop_T);

```

```

CL12(i) = Fz_T*cosd(a) - Fx_T*sind(a);
CD12(i) = Fz_T*sind(a) + Fx_T*cosd(a);

CL12_min(i) = min(CLs);
CL12_max(i) = max(CLs);
CD12_min(i) = min(CDs);
CD12_max(i) = max(CDs);
end

dCL12 = 0.5*(CL12_max-CL12_min);
dCD12 = 0.5*(CD12_max-CD12_min);
LD12 = CL12./max(CD12,eps);

% NACA 4412 Cp and uncert Cp
angles44 = [0 4 8 10 14];

Cp44_U = zeros(numel(angles44), numel(xC_4412_U));
Cp44_L = zeros(numel(angles44), numel(xC_4412_L));
dCp44_U = zeros(size(Cp44_U));
dCp44_L = zeros(size(Cp44_L));

for i = 1:numel(angles44)
    a = angles44(i);
    for p = 1:numel(xC_4412_U)
        hit = (aoa4412_U==a)&(port4412_U==p);
        if any(hit)
            pp = inH2O4412_U(hit)*INH2O_TO_PA; q=q4412_U(hit);
            Cp44_U(i,p) = pp./q;
            dCp44_U(i,p) = sqrt( (dp_scan./q).^2 + ((-pp./q.^2)).*dq_inf).^2 );
        end
    end
    for p = 1:numel(xC_4412_L)
        hit = (aoa4412_L==a)&(port4412_L==p);
        if any(hit)
            pp = inH2O4412_L(hit)*INH2O_TO_PA; q=q4412_L(hit);
            Cp44_L(i,p) = pp./q;
            dCp44_L(i,p) = sqrt( (dp_scan./q).^2 + ((-pp./q.^2)).*dq_inf).^2 );
        end
    end
end

% Force LE to 1
Cp44_U(:,1)=1;
Cp44_L(:,1)=1;

% NACA 0012 Cp plots
for i = 1:numel(angles44)
    U=Cp44_U(i,:);

```

```

L=Cp44_L(i,:);
dU=dCp44_U(i,:);
dL=dCp44_L(i,:);
cpTE = 0.5*(U(end)+L(end));
figure;
hold on;
errorbar([0 xC_4412_U 1],[1 U cpTE],[0 dU 0],'ko-', 'LineWidth',1.5,'DisplayName','Upper');
errorbar([0 xC_4412_L 1],[1 L cpTE],[0 dL 0],'ro-', 'LineWidth',1.5,'DisplayName','Lower');
xlabel('x/c');
ylabel('Cp');
title(sprintf('NACA 4412 Cp at %d (deg)',angles44(i)));
legend;
grid on;
axis ij;
end

```

```

% NACA 4412 Integrate CL and CD

```

```

% Closed loop geometry

```

```

xU = [0, xC_4412_U, 1];
zU = [0, zC_4412_U, 0];
xL = [0, xC_4412_L(end:-1:1), 1];
zL = [0, zC_4412_L(end:-1:1), 0];
xLoop44 = [xU, xL(2:end-1)];
zLoop44 = [zU, zL(2:end-1)];

```

```

CL44 = zeros(numel(angles44),1);
CD44 = zeros(numel(angles44),1);
CL44_min = CL44;
CL44_max = CL44;
CD44_min = CD44;
CD44_max = CD44;

```

```

for i = 1:numel(angles44)

```

```

    a = angles44(i);
    U_t = Cp44_U(i,:);
    l_t = Cp44_L(i,:);
    % bounds
    Umin = U_t - dCp44_U(i,:);
    Umax = U_t + dCp44_U(i,:);
    Lmin = l_t - dCp44_L(i,:);
    Lmax = l_t + dCp44_L(i,:);
    combos = { {Umin,Lmin},{Umin,Lmax},{Umax,Lmin},{Umax,Lmax} };
    CLs = zeros(1,4);
    CDs = zeros(1,4);
    for c = 1:4
        Uc=combos{c}{1};
        Lc=combos{c}{2};
        Uc(1)=1;
    end
end

```

```

    Lc(1)=1;
    cpTE = 0.5*(Uc(end)+Lc(end));
    cpU = [1, Uc, cpTE];
    cpL = [cpTE, Lc(end:-1:1), 1];
    cpLoop = [cpU, cpL(2:end-1)];
    Fx = trapz(zLoop44, cpLoop);
    Fz = trapz(xLoop44, -cpLoop);
    CLs(c) = Fz*cosd(a) - Fx*sind(a);
    CDs(c) = Fz*sind(a) + Fx*cosd(a);
end

U_t(1)=1;
l_t(1)=1;
cp_TEt = 0.5*(U_t(end)+l_t(end));
cp_Ut = [1, U_t, cp_TEt];
cp_Lt = [cp_TEt, l_t(end:-1:1), 1];
cp_Loop_T = [cp_Ut, cp_Lt(2:end-1)];
Fx_T = trapz(zLoop44, cp_Loop_T);
Fz_T = trapz(xLoop44, -cp_Loop_T);
CL44(i) = Fz_T*cosd(a) - Fx_T*sind(a);
CD44(i) = Fz_T*sind(a) + Fx_T*cosd(a);

CL44_min(i)=min(CLs);
CL44_max(i)=max(CLs);
CD44_min(i)=min(CDs);
CD44_max(i)=max(CDs);
end

dCL_44 = 0.5*(CL44_max-CL44_min);
dCD_44 = 0.5*(CD44_max-CD44_min);
LD_44 = CL44./max(CD44,eps);

%% Performance plots
maskPos12 = angles12>=0;

figure;
yyaxis left
errorbar(angles12(maskPos12),CL12(maskPos12),dCL12(maskPos12),'ro-', 'LineWidth',1.5,'DisplayName','C_L');
hold on;
errorbar(angles12(maskPos12),CD12(maskPos12),dCD12(maskPos12),'gs-', 'LineWidth',1.5,'DisplayName','C_D');
ylabel('Lift/Drag Coefficient');
yyaxis right
plot(angles12(maskPos12),LD12(maskPos12),'ko-', 'LineWidth',1.5,'DisplayName','L/D');
ylabel('L/D');
xlabel('Angle of Attack (deg)');
title('NACA 0012 C_L, C_D, L/D');
legend('Location','southeast');
grid on;

```

```
xlim([0 max(angles12(maskPos12))]);
```

```
figure;  
yyaxis left  
errorbar(angles44,CL44,dCL_44,'ro-','LineWidth',1.5,'DisplayName','C_L');  
hold on;  
errorbar(angles44,CD44,dCD_44,'gs-','LineWidth',1.5,'DisplayName','C_D');  
ylabel('Lift/Drag Coefficient');  
yyaxis right  
plot(angles44,LD_44,'ko-','LineWidth',1.5,'DisplayName','L/D');  
ylabel('L/D');  
xlabel('Angle of Attack (deg)');  
title('NACA 4412 C_L,C_D,L/D');  
legend('Location','southeast');  
grid on;
```

```
% uncert plots
```

```
figure; hold on;  
plot(angles12(maskPos12),CL12(maskPos12)+dCL12(maskPos12),'bo-','LineWidth',1.5,'DisplayName','C_L max');  
plot(angles12(maskPos12),CL12(maskPos12)-dCL12(maskPos12),'go-','LineWidth',1.5,'DisplayName','C_L min');  
plot(angles12(maskPos12),CD12(maskPos12)+dCD12(maskPos12),'bs-','LineWidth',1.5,'DisplayName','C_D max');  
plot(angles12(maskPos12),CD12(maskPos12)-dCD12(maskPos12),'gs-','LineWidth',1.5,'DisplayName','C_D min');  
plot(angles12(maskPos12),CL12(maskPos12),'k--o','DisplayName','C_L');  
plot(angles12(maskPos12),CD12(maskPos12),'k--s','DisplayName','C_D');  
xlabel('Angle of Attack (deg)');  
ylabel('Coefficient');  
title('NACA 0012 uncertainty');  
legend('Location','northwest');  
grid on;
```

```
figure; hold on;  
plot(angles44,CL44+dCL_44,'bo-','LineWidth',1.5,'DisplayName','C_L max');  
plot(angles44,CL44-dCL_44,'go-','LineWidth',1.5,'DisplayName','C_L min');  
plot(angles44,CD44+dCD_44,'bs-','LineWidth',1.5,'DisplayName','C_D max');  
plot(angles44,CD44-dCD_44,'gs-','LineWidth',1.5,'DisplayName','C_D min');  
plot(angles44,CL44,'k--o','DisplayName','C_L');  
plot(angles44,CD44,'k--s','DisplayName','C_D');  
xlabel('Angle of Attack (deg)');  
ylabel('Coefficient');  
title('NACA 4412 uncertainty');  
legend('Location','northwest');  
grid on;
```

```
% print results
```

```
% Labels
```

```
xlbl12 = ["LE","0.02","0.09","0.19","0.30","0.40","0.50","0.60","0.70","0.80","TE"];  
xlbl44U = ["LE","0.05","0.10","0.20","0.30","0.40","0.61","0.81","TE"];
```

```
xlbl44L = ["LE","0.10","0.20","0.30","0.40","0.55","0.70","TE"];
```

```
% NACA 0012 upper
fprintf('\nNACA 0012 upper surface Cp\n');
fprintf('Angle');
for j=1:numel(xlbl12), fprintf('%12s', xlbl12(j));
end
fprintf('\n');
fprintf(repmat('-',1,140)); fprintf('\n');
```

```
for i = 1:numel(angles12)
    a = angles12(i);
    if a < 0, continue;
    end
    cpU = Cp12(i,:);
    dU = dCp12(i,:);
    cpL_mate = Cp12(angles12== -a,:);
    dL_mate = dCp12(angles12== -a,:);
    if isempty(cpL_mate), cpL_mate=cpU;
        dL_mate=dU;
    end
    cpTE = 0.5*(cpU(end)+cpL_mate(end));
    dTE = 0.5*sqrt(dU(end).^2 + dL_mate(end).^2);
    fprintf('%3d°', a);
    fprintf(' %6.3f ± %-5.3f, 1.000, 0.000);
    for j=1:numel(cpU), fprintf(' %6.3f ± %-5.3f', cpU(j), dU(j));
    end
    fprintf(' %6.3f ± %-5.3f\n', cpTE, dTE);
end
```

```
% % NACA 0012 lower
fprintf('\nNACA 0012 lower surface Cp \n');
fprintf('Angle');
for j = 1:numel(xlbl12), fprintf('%12s', xlbl12(j));
end
fprintf('\n');
fprintf(repmat('-',1,140));
fprintf('\n');
```

```
posAngles12 = angles12(angles12 >= 0);
```

```
for i = 1:numel(posAngles12)
    aPos = posAngles12(i);

    % pull LOWER data from the -α row
    cp_Low = Cp12(angles12 == -aPos, :);
    d_Low = dCp12(angles12 == -aPos, :);
    if isempty(cp_Low)
```

```

        cp_Low = Cp12(angles12 == aPos, :);
        d_Low = dCp12(angles12 == aPos, :);
    end

    % mate UPPER row at  $+\alpha$ 
    cpUp = Cp12(angles12 == aPos, :);
    dUp = dCp12(angles12 == aPos, :);
    if isempty(cpUp)
        cpUp = cp_Low;
        dUp = d_Low;
    end

    % TE value and uncertainty
    if aPos == 0
        cpTE = cp_Low(end);
        dTE = d_Low(end);
    else
        cpTE = 0.5*(cp_Low(end) + cpUp(end));
        dTE = 0.5*sqrt(d_Low(end).^2 + dUp(end).^2);
    end

    fprintf('%3d°', aPos);
    fprintf(' %6.3f ± %-5.3f, 1.000, 0.000);
    for j = 1:numel(cp_Low)
        fprintf(' %6.3f ± %-5.3f, cp_Low(j), d_Low(j));
    end
    fprintf(' %6.3f ± %-5.3f\n', cpTE, dTE);
end

% NACA 4412 upper
fprintf('\nNACA 4412 UPPER Cp (inline uncertainty: value ±  $\delta$ )\n');
fprintf('Angle');
for j=1:numel(xlbl44U), fprintf('%12s', xlabel44U(j));
end
fprintf('\n');
fprintf(repmat('-',1,108));
fprintf('\n');

for i=1:numel(angles44)
    U=Cp44_U(i,:);
    dU=dCp44_U(i,:);
    L=Cp44_L(i,:);
    dL=dCp44_L(i,:);
    cpTE=0.5*(U(end)+L(end));
    dTE=0.5*sqrt(dU(end).^2 + dL(end).^2);
    fprintf('%3d°', angles44(i));
    fprintf(' %6.3f ± %-5.3f, 1.000, 0.000);
    for j=1:numel(U), fprintf(' %6.3f ± %-5.3f, U(j), dU(j));

```

```

    end
    fprintf(' %6.3f ± %-5.3f\n', cpTE, dTE);
end

% NACA 4412 lower
fprintf('\nNACA 4412 LOWER Cp (inline uncertainty: value ± δ)\n');
fprintf('Angle');
for j=1:numel(xlbl44L), fprintf('%12s', xlabel44L(j));
end
fprintf('\n');
fprintf(repmat('-',1,100));
fprintf('\n');

for i=1:numel(angles44)
    U=Cp44_U(i,:);
    dU=dCp44_U(i,:);
    L=Cp44_L(i,:);
    dL=dCp44_L(i,:);
    cpTE=0.5*(U(end)+L(end));
    dTE=0.5*sqrt(dU(end).^2 + dL(end).^2);
    fprintf('%3d°', angles44(i));
    fprintf(' %6.3f ± %-5.3f, 1.000, 0.000);
    for j=1:numel(L), fprintf(' %6.3f ± %-5.3f, L(j), dL(j));
    end
    fprintf(' %6.3f ± %-5.3f\n', cpTE, dTE);
end

% C_L,C_D,L/D tables
fprintf('\nNACA 0012 C_L,C_D,L/D\n');
fprintf('Angle C_L C_D L/D\n');
fprintf('-----\n');
for i=1:numel(angles12)
    a=angles12(i);
    if a<0, continue;
    end
    fprintf('%2d° %7.4f ± %-6.4f %7.4f ± %-6.4f %6.2f\n', ...
        a, CL12(i), dCL12(i), CD12(i), dCD12(i), LD12(i));
end

fprintf('\nNACA 4412 C_L,C_D,L/D\n');
fprintf('Angle C_L C_D L/D\n');
fprintf('-----\n');
for i=1:numel(angles44)
    fprintf('%2d° %7.4f ± %-6.4f %7.4f ± %-6.4f %6.2f\n', ...
        angles44(i), CL44(i), dCL_44(i), CD44(i), dCD_44(i), LD_44(i));
end

```


VII. Acknowledgements

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